

JKKN COLLEGE OF ARTS & SCIENCE
(AUTONOMOUS)
(ACCREDITED BY NAAC, APPROVED BY AICTE,
AFFILIATED TO PERIYAR UNIVERSITY, SALEM-11)
KOMARAPALAYAM- 638 183, NAMAKKAL (Dt).



BACHELOR OF COMPUTER APPLICATIONS PROGRAMME

OBE UNDER CHOICE BASED CREDIT SYSTEM
(With effect from the academic year 2025-2026 onwards)

Introduction

BACHELOR OF COMPUTER APPLICATIONS

Education is the key to development of any society. Role of higher education is crucial for securing right kind of employment and also to pursue further studies in best available world class institutes elsewhere within and outside India. Quality education in general and higher education in particular deserves high priority to enable the young and future generation of students to acquire skill, training and knowledge in order to enhance their thinking, creativity, comprehension and application abilities and prepare them to compete, succeed and excel globally. Learning Out comes-based Curriculum Framework (LOCF) which makes it student-centric, interactive and outcome-oriented with well-defined aims, objectives and goals to achieve. LOCF also aims at ensuring uniform education standard and content delivery across the state which will help the students to ensure similar quality of education irrespective of the institute and location.

Computer Application is the study of quantity, structure, space and change, focusing on problem solving, application development with wider scope of application in science, engineering, technology, social sciences etc. throughout the world in last couple of decades and it has carved out a space for itself like anyother disciplines of basic science and engineering. Computer Application is a discipline that spans theory and practice and it requires thinking both in abstract terms and in concrete terms.

Nowadays, practically everyone is a computer user, and many people are even computer programmers. Computer Application can be seen on a higher level, as a science of problem solving and problem solving requires precision, creativity, and careful reasoning. The ever-evolving discipline of computer Application also has strong connections to other disciplines. Many problems in science, engineering, health care, business, and other areas can be solved effectively with computers, but finding a solution requires both computer science expertise and knowledge of the particular application domain. Computer Application has a wide range of specialties. These include Computer Architecture, Software Systems, Graphics, Artificial Intelligence, Computational Science, and Software Engineering. Drawing from a common core of computer science knowledge, each specialty area focuses on specific challenges. Computer Application is practiced by mathematicians, scientists and engineers. Mathematics, the origins of Computer Science, provides reason and logic. Science provides the methodology for learning and refinement. Engineering provides the techniques for building hardware and software.

I) Definitions:

Programme : A course of study leading to the award of a degree in a discipline.

Course : Course refers to the subject offered under the degree programme.

Part I : Tamil/other languages

Part II : English language

Part III : (1) The core courses related to the programme concerned and practical.
(2) Elective generic/discipline specific courses related to the core courses of the programme concerned.

Part IV:

- ❖ Non-Major Elective means, elective subjects offered by other Major Departments and are to be selected by the students.
- ❖ Skill Enhancement Courses means the courses offered as skill based courses aimed at imparting Advanced Skills.
- ❖ Value Education (offered in the III year - V Semester) and Environmental Studies (offered in the II year - III & IV Semesters)
- ❖ Foundation course is a paper to ease the transition of learning from higher secondary to higher education, providing basic concepts in the particular course.
- ❖ The students have to undergo Internship for a minimum period of 15 days and maximum 30 days at the end of semester I, III, V during the vacation. At the end of the respective (II, IV, VI Semester), Viva - Voce will be conducted on the basis of the report submitted by the students. Viva-Voce shall be conducted by the Internal / External Examiner.

Part V:

Extension Activities: All Outreach Programmes, Co-Curricular and Extracurricular Activities.

A detailed explanation of the above with relevant credits are given under Scheme of Examination along with Distribution of Marks

Duration: It refers to the stipulated years of study to complete a programme as prescribed by the University from time to time. Currently for the undergraduate programme the duration of study is THREE years. These regulations shall apply to the regular course of study in approved institutions of the University.

Credits: It refers to the weightage or points assigned to each course of study (subject) by the experts of the respective Board of Studies.

Credit System: It refers to the course of study under this regulation, where weightage of credits and the Cumulative Grade Point is spread over to different semesters during the period of study. Average grade point shall be calculated based on the credits earned by the students. A total of 140 credits with 4 extra credits is recommended for the Undergraduate Programme (Respective Board).

Choice-Based Credit System:

All Undergraduate Programmes offered by the College shall be under Choice Based Credit System (CBCS). This is to enhance the quality and mobility of the students within and between the Universities in the country and abroad.

II) Eligibility for Admission to the Course:

A candidate who has passed in Higher Secondary Examination with Mathematics or Business Mathematics or Computer Science or Statistics (Academic stream or Vocational stream) as one of the subject under Higher Secondary Board of Examination, Tamilnadu as per norms set by the Government of Tamilnadu or an Examination accepted as equivalent there to.

III) Duration of the Course:

The course shall extend over a period of three years comprising six semesters with two semesters in one academic year. There shall not be less than 90 working days for each semester. Examination shall be conducted at the end of every semester for the respective subjects.

IV) CBCS SYSTEM:

The UG programme shall be conducted on Choice Based Credit System (CBCS). The term credit refers to the weightage given to a course. However, in no instance the credit of a course can be greater than the hours allotted to it. Each Course is designed variously under Lectures / Tutorials / Laboratory or Field work /Seminar / Practical training / Assignments / Report writing etc., to meetthe effective teaching and learning needs.

V) Course of Study:

The course of study for the UG degree courses of all branches consists the following:

Part - I:

Tamil or any one of the following modern/classical languages i.e. Telugu, Kannada, Malayalam, Hindi, Sanskrit, French, German, Arabic & Urdu.

The subject shall be offered during I, II, III, IV semesters with one examination at the end of each semester (12 credits).

Part – II: English

The subject shall be offered during I, II, III, IV semesters with one examination at the end of each semester (12 credits).

Part – III:

a) Core subjects:

As prescribed in the scheme of examination.

b) Elective courses:

Elective subjects are to be selected from the list of generic/discipline specific electives prescribed by the respective Board of Studies.

Part – IV :Comprises of 1) Skill Enhancement courses, 2) Environmental Studies, 3) Value Education, 4) Non major elective, 5) Foundation course, 6) Internship / Industrial training, 7) Professional Competency Skill.

Part – V: Extension Activities

Every student must compulsorily participate during the specified period in any one of the Outreach Programmes, including Co-Curricular or Extra-Curricular activities.

VI) CLASSIFICATION OF THE SUCCESSFUL CANDIDATES:

Candidates who obtain 75% and above in the aggregate shall be deemed to have passed the examination in First Class with Distinction provided they pass all the examinations prescribed for the programme at the first appearance. Candidates, other than the above, who secure not less than 60% of the aggregate marks in the whole examinations, shall be declared to have passed the examination in First Class. The candidates who secure 50% to 59.99 % in the aggregate shall be declared to have passed in Second class. The candidates who secure 40% to 49.99 % in the aggregate shall be declared to have passed in Third class. The candidate who secures less than 40% is declared as Re-appear. He/she can appear in the forthcoming semester examination.

Candidates who pass all the examinations prescribed for the programme in the first instance and within a period of three academic years from the year of admission are only eligible for Ranking.

VII) Examination:

To be conducted at the end of each semester. For Environmental Studies and Practical refer Scheme of Examination.

VIII) Eligibility to appear for the examinations:

A Candidate shall be permitted to appear for the end semester examinations for any semester (practical/theory) if he/she secures not less than 75% of attendance in the number of working days during the semester.

A candidate who secures attendance from 65% to 74.99% has to pay a fine of Rs.800/- to appear for the examination.

A candidate who secures attendance from 50 % to 64.99% has to pay a fine of Rs.800/- and can appear for both semester papers together at the end of the later semester.

A candidate who has secured less than 50% of attendance in any semester shall not be permitted to appear for the regular examinations and has to continue the study in the subsequent semester. He/she has to re-join the semester in which the attendance is less than 50%.

IX) Restrictions to appear for the examinations:

Any candidate declared as re-appear shall have the option to re-appear along with the regular semester papers. Candidate who is declared as re-appear in any of the course of Part I, II, III, IV & V of UG degree examinations shall complete the course concerned within 5 years from the date of admission to the said programme and if they fail to do so, shall take up the examination in the revised syllabus prescribed for the immediate next batch of candidates.

If there is no change in the syllabus they shall appear for the examination in that course with the syllabus in vogue until there is a change in the syllabus.

In the event of removal of that course, after 5-years period, the candidates shall have to take up an equivalent course in the revised syllabus as suggested by the Chairman of the respective Board and fulfil the requirements as per the regulations of the curriculum for the award of the degree.

X) Medium of Instruction and examinations:

The medium of instruction and examinations for the courses of Part I & II shall be the language concerned. For part III & IV, the medium of instruction is English.

XI) Submission of Record Note Books for practical examinations:

Candidates appearing for practical examinations should submit bonafide record note books prescribed for practical examinations, otherwise the candidates shall not be permitted to appear for the practical examinations.

XII) Passing Minimum & Mark Distribution:

The following are the distribution of marks for **End Semester** (External) examination and continuous internal assessment and passing minimum marks for theory papers of UG programmes.

a. End Semester Examination

Total Marks	External		Internal	Overall Passing Minimum for total marks (Internal + External)
	Max. Marks	Passing Minimum for external alone	Max. Marks	
100	60	24	40	40

b. The following are the distribution of marks for continuous internal assessment in theory papers of UG programmes:

Test No.	Max Marks	Weightage
CIA-I	30	10
CIA-II	30	
Model examination	60	10
Seminar, Class Activities	10	10
Attendance	5	5
Assignment/Project	5	5
Total		40

c. Mark Allotment for Practical \ Internship

Category	Internal	External	Total Marks	Passing minimum
Practical	40	60	100	40 marks (40%)
Internship	Commended/Highly Commended			

- d. The following are the distribution of marks for the Continuous Internal Assessment in UG Practical Courses:

S.No.	For Practical – UG Courses	Distribution of Marks
1	Observation Record	10
2	Model Examination	15
3	Record	10
4	Attendance	5
Total		40

The following are the distribution of marks for End Semester Examination in Practical Papers of UG Programmes:

Category of Practical External	Mark split-up (60 marks)
Algorithm/Procedure/Flowchart	5 Marks
Program Coding / Experimental Setup	10 Marks
Execution / Output Verification	10 Marks
Viva-Voce (related to the question)	5 Marks
Total per Question	30 Marks

Question Format: There will be TWO Questions:

Question No.	Type	Marks
Q1	Program/Experiment based on core concepts	30 Marks
Q2	Program/Experiment based on core concepts	30 Marks

→ Students must attempt BOTH questions.

XIII)a) TEST PORTION ALLOCATION FOR CIA & MODEL EXAMINATIONS:

Test No.	Portion	Total duration	Max. marks
CIA-I	2 units	1 hour	30
CIA-II	2 units	1 hour	30
Model Examination	5 units	3 hours	60

b) WEIGHTAGE OF INTERNAL MARKS:

Test No.	Max Marks	Weightage
CIA-I	30	10
CIA-II	30	
Model examination	60	10
Seminar, Class Activities	10	10
Attendance	5	5
Assignment/Project	5	5
Total		40

- ❖ **Class Activity:** Flipped classroom: Assignment / Case study / Field survey / PPT presentation / group discussion / Subject Viva-voce / Report Writing / Content Writing / Survey studies / Mind map / Flow chart preparation / Model making / Review of report / Debate / Group Discussion / Surprise test / Open book test etc.

XIV)

**a. QUESTION PAPER PATTERN FOR CONTINUOUS INTERNAL ASSESSMENT
(30 Marks)**

Section	Fink's Taxonomy	Marks	Total
A (4 Questions X 1 Mark)	K1, K2 K3, K4 K5, K6	4	30
B (2 Questions X 5 Marks) Open choice question		10	
C (2 Question X 8 Marks) Internal choice (Either or pattern)		16	

b) QUESTION PAPER PATTERN FOR END SEMESTER EXAMINATIONS (60 MARKS)

Section	Question Type	Fink's Dimension	Marks Description
A	Multiple Choice Questions (10 × 1 marks)	Foundational Knowledge	10
B	Open Choice Question (2 × 5 marks)	Application	10
C	Internal choice (Either or pattern) (5 x 8 marks)	Integration + Learning How to Learn	60
Total			40 Marks

K1-Foundational Knowledge
K2-Application
K3-Integration

K4-Human Dimension
K5-Caring
K6-Learning How to Learn

XV) BREAKUP OF MARKS FOR ATTENDANCE:

The students are awarded marks for their attendance by following the breakups mentioned below

Percentage	Marks
90 and above	- 5
85 - 89.99	- 4
75 - 84.99	- 3
Less than 75	- 2

As a special case under Medical / Sports / other activities, attendance exemption may be given by the Principal.

XVI) Continuous Internal Assessment shall be conducted by the COE and the marks shall be submitted to the COE by the respective departments.

XVII) A Candidate who passes the examination in all the courses of Part I, II, III, IV& V shall be declared to have passed the whole examination.

XVIII) The syllabus for various courses shall be clearly demarcated into five viable units in each paper/subject.

XIX) Revision of Regulations and Curriculum:

The above Regulation and Scheme of Examinations shall be in vogue without any change for a minimum period of three years from the date of approval. The Board of Studies may revise / amend the Regulations and Scheme of Examinations, if found necessary.

XX) Internship (Minimum of 30 days):

The students have to undergo Internship for a minimum period of 30 days at the end of semester IV during the vacation. They have the option to select any organization – Government / private like Industry, R & D Organizations, Scientific companies, etc., in consultation with the staff coordinator & Head of the Department. They must maintain a work diary and prepare a report of the training undergone and submit the same. At the end of the V Semester, Viva -Voce will be conducted on the basis of the report submitted by the students. Viva-Voce shall be conducted by the Internal Examiner.

XXI) Project work:

UG Science:

The Marks for the Project work is 100.

Internal Assessment: 40 Marks (2 Reviews 20+20)

Viva voce: 20 Marks

Evaluation of Project: 40 Marks

GUIDELINES FOR ASSESSMENT:

- The program for the degree of Bachelor of Science in Computer Science consists of three academic years and divided into six semesters.
- Courses and their credit values will be decided by the respective board.
- The regulations shall take effect from the academic year 2025-26, i.e., for students who are to be admitted in the first year of the program during the academic year 2025-26 and thereafter.
- The theory examinations will be conducted for three hours duration to each paper at the end of each semester.
- The Candidate declared reappear in any subject(s) will be permitted to appear for each failed subject(s) in the subsequent examinations.
- Marks for theory/practical/project courses shall be out of 100.
- The learners must complete a minimum of 30 days Internship at the end of the fifth semester examinations. Summer Internship course will be assessed by **Commended or Highly commended**.
- The Extension Activity shall be completed in the sixth semester. Assessment of this activity is done by providing only one credit No marks are awarded.
- Outreach Programmes, including Co-Curricular or Extra-Curricular activities are considered as extension activities.
- If a candidate fails to submit the report on project work on or before the date specified by the Controller of Examinations, he/she shall have reappeared and can submit the report for evaluation in the upcoming semester itself.
- A total of 140 credits with 4 extra credits is recommended for the Undergraduate Programme (Respective Board).
- The final assessments will be based on summing up of Internal and End Semester Examination.
- If a student is disabled, a scribe from other discipline can be arranged by the department with the concurrence of the Chief Superintendent.

Credit distribution for Bachelor of computer applications under Learning Outcome based Curriculum framework –Choice Based Credit System Pattern with effect from 2025-2026 Onwards.

XXII) Credit Distribution Based on TANSCH/UGC Guidelines

Part	Subjects	No. of Credits
PART-I	Language	12
PART-II	English	12
PART-III	Core Courses & Elective Courses[in Total]	87
PART-IV	Skill Enhancement Courses	12
	Environmental Studies	2
	Value Education	2
	Non Major Elective	8
	Internship / Industrial Training	6
	Professional Competency Skill	2
PART-V	Extension Activity	1
Grand Total		
Extra Credit (if any)		
	Total No. of Subjects	43
	Total credits	144

XXIII) Consolidated Semester wise and Component wise Credit distribution.

Parts	Sem I	Sem II	Sem III	Sem IV	Sem V	Sem VI	Total Credits
Part I	3	3	3	3	-	-	12
Part II	3	3	3	3	-	-	12
Part III	11	11	12	13	22	18	84
Part IV	6	6	7	8	2	4	35
Part V	-	-	-	-	-	1	1
Total	23	25	24	25	24	23	144

*Part I, II, and Part III components will be separately taken into account for CGPA (Cumulative Grade Point Average) calculation and classification for the under graduate programme and the other components. IV, V have to be completed during the duration of the programme as per the norms, to be eligible for obtaining the UG degree.

XXIV) Grading

The CIA and End semester examination marks will be taken into account for grading.

The following table gives the marks, grade points, letter grades and classification to indicate the performance of the candidate.

Range of marks	Grade points	Letter grade	Description
90-100	9.0-10.0	O	OUTSTANDING
80-89	8.0-8.9	D+	EXCELLENT
75-79	7.5-7.9	D	DISTINCTION
70-74	7.0-7.4	A+	VERY GOOD
60-69	6.0-6.9	A	GOOD
50-59	5.0-5.9	B	AVERAGE
40-49	4.0-4.9	C	SATISFACTORY
00-39	0.0	U	RE-APPEAR
Absent	0.0	AAA	ABSENT

Grade point average (for a Semester):

C_i = Credits earned for course / in any semester.

G_i = Grade Point obtained for course / in any semester.

n = refers to the semester in which such course was credited.

Calculation of grade point average semester-wise and part-wise is as follows:

$$\text{GRADE POINT AVERAGE [GPA]} = \frac{\sum C_i G_i}{\sum C_i}$$

$$\text{GPA} = \frac{\text{Sum of the multiplication of grade points by the credits of the courses offered under each part}}{\text{Sum of the credits of the courses under each part in a semester}}$$

Calculation of Grade Point Average (CGPA) (for the entire programme):

A candidate who has passed all the examinations under different parts (Part-I to V) is eligible for the following part-wise computed final grades based on the range of CGPA:

$$\text{Cumulative Grade Point Average [CGPA]} = \frac{\sum C_i G_i}{\sum C_i}$$

$$\text{CGPA} = \frac{\text{Sum of the multiplication of grade points by the credits of the entire programme under each part}}{\text{Sum of the credits of the course of the entire programme under each part}}$$

For the entire programme:

CGPA	GRADE	CLASSIFICATION OF FINAL RESULT
9.5-10.0	O+	First Class - Exemplary*
9.0 and above but below 9.5	O	
8.5 and above but below 9.0	D++	
8.0 and above but below 8.5	D+	
7.5 and above but below 8.0	D	
7.0 and above but below 7.5	A++	First Class
6.5 and above but below 7.0	A+	
6.0 and above but below 6.5	A	
5.5 and above but below 6.0	B+	
5.0 and above but below 5.5	B	Second Class
4.5 and above but below 5.0	C+	
4.0 and above but below 4.5	C	

*The Candidates who have passed in the first appearance and within the prescribed semester of the UG programme are eligible.

XXV) Credit Distribution for UG Programme Semester wise

Sem I	Credit	H	Sem II	Credit	H	Sem III	Credit	H	Sem IV	Credit	H	Sem V	Credit	H	Sem VI	Credit	H
Part 1. Language – Tamil	3	5	Part..1. Language – Tamil	3	5	Part..1. Language – Tamil	3	5	Part..1. Language – Tamil	3	5	5.1 Core Course –\CC IX	4	5	6.1 Core Course – CC XIII	4	6
Part.2 English	3	5	Part..2 English	3	5	Part..2 English	3	5	Part..2 English	3	5	5.2 Core Course – CC X	4	5	6.2 Core Course – CC XIV	4	6
1.3 Core Course – CC I	5	5	2..3 Core Course – CC III	5	6	3.3 Core Course – CC V	5	5	4.3 Core Course – CC VII - Core Industry Module	5	6	5. 3.Core Course CC – XI	4	5	6.3 Core Course – CC XV	4	6
1.4 Core Course – CC II	3	4	2.4 Core Course – CC IV	3	3	3.4 Core Course – CC VI	3	3	4.4 Core Course – CC VIII	3	3	5. 4.Core Course –/ Project with viva- voce CC -XII	3	4	6.4 Elective -VII Generic/ Discipline Specific	3	5
1.5 Elective I Generic/ Discipline Specific	3	5	2.5 Elective II Generic/ Discipline Specific	3	5	3.5 Elective III Generic/ Discipline Specific	3	5	4.5 Elective IV Generic/ Discipline Specific	3	5	5.5 Elective V Generic/ Discipline Specific	3	4	6.5 Elective VIII Generic Discipline Specific	3	5
1. 6 NME - I	2	2	2.6 NME - II	2	2	3.6 NME-III	2	2	4.6 NME-VI	2	2	5.6 Elective VI Generic/ Discipline Specific	4	5	6.6 Extension Activity	1	-
1.7 Skill Enhancement Course –SEC-I	4	4	2.7 Skill Enhancement Course – SEC-II	4	4	3.7 Skill Enhancement Course SEC- III (Entrepreneurial Skill)	4	4	4.7 Skill Enhancement Course SEC-V	4	4	5.7 Value Education	2	2	6.7 Professional Competency Skill Enhancement	2	2
			2.8 Summer Internship /Industrial Training	2		3.8 EVS	1	1	4.8 Summer Internship /Industrial Training	2				-	6.8 Summer Internship /Industrial Training	2	
	23	30		25	30		24	30		25	30		24	30		23	30
Total – 144 Credits																	

XXVI)

a. Methods of Assessment:

Levels	Methods of Assessment
Foundational Knowledge (K1)	Simple definitions, MCQ, Recall steps, Concept definitions
Application(K2)	MCQ, True/False, Short essays, Concept explanations, Short summary or overview
Integration(K3)	Suggest idea/concept with examples, Suggest formulae, Solve problems, Observe, Explain
Human Dimension (K4)	Problem-solving questions, Finish a procedure in many steps, Differentiate between various ideas, Map knowledge
Caring (K5)	Longer essay/Evaluation essay, Critique or justify with pros and cons
Learning How to Learn (K6)	Check knowledge in specific or off beat situations, Discussion, Debating or Presentations

b. Methods of Evaluation:

METHODS OF EVALUATION		
Internal Evaluation	Formative (20)	40 Marks
	Summative (20)	
External Evaluation	End Semester Examination	60 Marks
	Total	100Marks

XXVII)

a. Mandatory Core Areas for B.SC CS Programme

I Year	Sem-I	25UCAC01	Core - I - Python Programming
		25UCACP01	Core Practical – I- Python Programming Lab
	Sem-II	25UCAC02	Core – II - Object Oriented Programming using C++
		25UCACP03	Core Practical – II - C++ Programming Lab

b. Mandatory Generic Electives for B.SC CSProgramme

I Year	Sem-I	25UMAGE3 /25UMAGE6 /25UMAGE7 /25UMAGE9	Generic Elective – I Discrete Mathematics-I / Optimization Techniques / Introduction to Linear Algebra / Numerical Methods-I
	Sem-II	25UMAGE4 / 25UMAGE5 / 25UMAGE8/ 25UMAGE10	Generic Elective– II Discrete Mathematics-II /Numerical Methods / Graph Theory and its Applications /Numerical Methods-II
		25UMAGEP02 / 25UMAGEP03 / 25UMAGEP04 / 25UMAGEP05 / 25UMAGEP06	Generic Elective - Practical – I Discrete Mathematics / Numerical Methods / Optimization Techniques / Introduction Linear Algebra / Graph Theory and its Applications

c. NME (For other major students)

I Year	Sem-I	25UCANM1	NME - I - Digital advertising and strategies
	Sem-II	25UCANM2	NME - II - Digital skills for Employability

d. SEC

I Year	Sem-I	25UCAS01	SEC - I – Computer Science with AI Acceleration - Foundations
	Sem-II	25UCAS02	SEC - II – Computer Science with AI Acceleration - Implementation

Programme Outcomes:

PO1: Foundational Knowledge & Practical Skills:

Develop scientific aptitude and acquire basic practical skills in computer science and related fields; Gain domain knowledge in various computer science and humanities subjects; and possess the basic subject knowledge required for higher studies and professional courses.

PO2: Career & Further Education:

Become highly employable in the education sector, industry, or pursue entrepreneurial endeavours.

PO3: Problem-Solving & Social Responsibility:

Develop a solution-oriented approach towards social and environmental issues; and Utilize analytical reasoning skills to model and solve real-life problems.

PO4: Analytical & Communication Skills:

Enhance proficiency in analytical reasoning through critical understanding, analysis and synthesis; and Hone your ability to recognize patterns, identify key aspects of problems and share insights effectively.

PO5: Responsible Global Citizen:

Cultivate the skills and knowledge to become a responsible citizen in a rapidly changing and interconnected world.

Programme Specific Outcomes

PSO1: Think in a critical and logical based manner

PSO2: Capacity to transfer the skills gained, to provide innovative and novel solutions by maintaining ethical norms for the betterment of society.

PSO3: Familiarize the students with suitable software tools of computer science and industrial applications to handle issues and solve problems in mathematics or statistics and real-time application related sciences.

PSO4: Equip with Computer science technical ability, problem solving skills, creative talent and power of communication necessary for various forms of employment.

PSO5: Acquire good knowledge and understanding to solve specific theoretical and applied problems in advanced areas of Computer Science and Industrial statistics.

Mapping of Course Outcomes (COs) with Programme Outcomes (POs) and Programme Specific Outcomes (PSOs) can be carried out accordingly, assigning the appropriate level in the grids :(put tick mark in each row)

PO/PSO	PSO1	PSO2	PSO3	PSO4	PSO5
PO1	✓				
PO2		✓			
PO3			✓		
PO4				✓	
PO5					✓

BCA Programme:
SCHEME OF LEARNING AND EVALUATION
 (For the students admitted from the academic year 2025 – 2026 onwards)
 Outcome Based Curriculum Framework with CBCS

FIRST SEMESTER										
Part	Course Code	Title of the Course	Exam (Hrs)	Credits	Hours			Max. Marks		
					L	P	Total	CIA	ESE	Total
Part- I	25UGTA01	General Tamil-I	3	3	5	-	5	40	60	100
Part- II	25UGEN01	General English-I	3	3	5	-	5	40	60	100
Part- III	25UCAC01	Core - I - Python Programming	3	5	6	-	5	40	60	100
	25UCACP01	Core Practical – I- Python Programming Lab	3	3	-	4	4	40	60	100
	25UMAGE	Generic Elective-I	3	5	5	-	5	40	60	100
Part- IV	25UCANM1	NME - I - Digital Advertising and strategies	3	2	2	-	2	40	60	100
	25UCAS01	SEC – I Computer Science with AI Acceleration –foundations	3	4	4	-	4	40	60	100
	TOTAL			23	26	4	30			700
SECOND SEMESTER										
Part- I	25UGTA02	General Tamil-II	3	3	5		5	40	60	100
Part- II	25UGEN02	General English-II	3	3	5		5	40	60	100
Part- III	25UCAC02	Core – II - Object Oriented Programming using C++	3	5	5		5	40	60	100
	25UCACP03	Core Practical – II - C++ Programming Lab	3	3	-	4	4	40	60	100
	25UMAGE	Generic Elective-II	3	3	5	-	5	40	60	100
Part- IV	25UCANM2	NME- II - Digital skills for Employability	3	2	2	-	2	40	60	100
	25UCAS02	SEC – II - Computer Science with AI Acceleration – Implementation	3	4	4	-	4	40	60	100
		Summer Internship /Industrial Training		2						
	TOTAL			25	26	4	30			800

Signature of the Board Chairman

Signature of the CDC Co-ordinator

Signature of the Principal

Annexure – I: Elective Course

Generic Elective Specific

S.No	Paper code	Paper Title
1		Mathematics-I
2		Mathematics-II
3		Mathematics Practical
4		Discrete Mathematics-I
5		Discrete Mathematics-II
6		Discrete Mathematics Practical
7		Numerical Methods
8		Optimization Techniques
9		Optimization Techniques Practical
10		Introduction to Linear Algebra
11		Introduction to Linear Algebra Practical
12		Graph Theory and its Application
13		Graph Theory and its Application Practical
14		Numerical Methods-I
15		Numerical Methods-II
16		Numerical Methods Practical
17		Statistical Methods and its Application-I
18		Statistical Methods and its Application-II
19		Statistical Practical
20		Physics-I
21		Physics Practical-I
22		Physics-II
23		Physics Practical-II
24		Digital Logic Fundamentals
25		Nano Technology
26		Resource Management Techniques and more

Discipline Elective Specific

S.No	Paper Code	Paper Title
1	25UCADE01	Natural Language Processing
2	25UCADE02	Analytics for Service Industry
3	25UCADE03	Cryptography
4	25UCADE04	Big Data Analytics
5	25UCADE05	IOT and its Applications

6	25UCADE06	Software Project Management
7	25UCADE07	Image Processing
8	25UCADE08	Human Computer Interaction
9	25UCADE09	Fuzzy Logic
10	25UCADE10	Artificial Intelligence
11	25UCADE11	Robotics and its Applications
12	25UCADE12	Computational Intelligence
13	25UCADE13	Grid Computing
14	25UCADE14	Cloud Computing
15	25UCADE15	Artificial Neural Network
16	25UCADE16	Introduction to Data Science
17	25UCADE17	Agile Project Management
18	25UCADE18	Virtual Reality and more

Skill Enhancement Course (SEC)

PART	SEM	SUBJECT CODE	SUBJECT
IV	I	25UCAS01	SEC - I – Computer Science with AI Acceleration - Foundations
	II	25UCAS02	SEC - II – Computer Science with AI Acceleration - Implementation

Non Major Elective – (NME)

Subjects offered by the Department of Computer Science

The department offer the subjects to the other major subject students in each semester.

PART	SEM	SUBJECT CODE	SUBJECT
IV	I	25UCANM1	NME - I - Digital advertising and strategies
	II	25UCANM2	NME - II - Digital skills for Employability
	III	25UCANM3	NME -III - Website Optimization for marketing
	IV	25UCANM4	NME- IV - Office Management

SEMESTER - I

Core-I	PYTHON PROGRAMMING		
Course code	Total Hours	Contact Hours	Credits
25UCAC01	5	5	5
	Course Objectives		
	The main objectives of this course are		
1	To make students understand the concepts of Python programming.		
2	To apply the OOPs concept in PYTHON programming.		
3	To impart knowledge on demand and supply concepts		
4	To make the students learn best practices in PYTHON programming		
5	To know the costs and profit maximization		
CO numbers	Expected Course Outcome: On the successful completion of the course, student will be able to:		
CO1	To make students understand the concepts of Python programming.	K1 K2 K3 K2 K4 K5 K6	
CO 2	To apply the OOPs concept in PYTHON programming.		
CO 3	To impart knowledge on demand and supply concepts.		
CO 4	To make the students learn best practices in PYTHON programming		
CO 5	To know the costs and profit maximization.		
	K1 -Foundational Knowledge, K2 -Application, K3 -Integration, K4 -Human Dimension, K5 -Caring, K6 -Learning How to Learn		
	Course content		
I	Python Programming: History of Python-Features of Python-Setting up Python development environment (IDEs, Jupyter Notebooks, etc.), -Literal-Constants-Variables - Identifiers–Keywords-Built-in Data Types-Output Statements – Input Statements-Comments – Indentation- Operators-Expressions- Type conversions. Python Arrays: Defining and Processing Arrays – Array methods.		
II	Control Statements: Selection/Conditional Branching statements: if, if-else, nested if and if-elif-else statements. Iterative Statements: while loop, for loop, else suite in loop and nested loops. Jump Statements: break, continue and pass statements.		
III	Functions: Function Definition – Function Call – Variable Scope and its Lifetime-Return Statement. Function Arguments: Required Arguments, Keyword Arguments, Default Arguments and Variable Length Arguments- Recursion. Python Strings: String operations- Immutable Strings - Built-in String Methods and Functions - String Comparison. Modules: import statement- The Python module – dir() function – Modules and Namespace – Defining our own modules.		
IV	Lists: Creating a list -Access values in List-Updating values in Lists-Nested lists - Basic list operations-List Methods. Tuples: Creating, Accessing, Updating and Deleting Elements in a tuple – Nested tuples– Difference between lists and tuples. Dictionaries: Creating, Accessing, Updating and Deleting Elements in a Dictionary – Dictionary Functions and Methods - Difference between Lists and Dictionaries.		
V	Python File Handling: Types of files in Python - Opening and Closing files-Reading and Writing files: write() and writelines() methods- append() method – read() and		

	readlines() methods – with keyword – Splitting words – File methods - File Positions- Renaming and deleting files. *Case Study: ❖ Python using AI Object-oriented programming concepts, Numpy ,File Handling (working with CSV, XML and JSON files) , Jupyter Notebook, PyCharm, etc. (Student must be submit the report)		
Text Books	1. ReemaThareja, —Python Programming using problem solving approach, First Edition, 2017, Oxford University Press. 2. Dr. R. NageswaraRao, —Core Python Programming, First Edition, 2017, Dream tech Publishers.		
Reference Books	1. VamsiKurama, —Python Programming: A Modern Approach, Pearson Education. 2. Mark Lutz, Learning Python, Orielly. 3. Adam Stewarts, —Python Programming, Online. 4. Fabio Nelli, —Python Data Analytics, APress. 5. Kenneth A. Lambert, —Fundamentals of Python – First Programs, CENGAGE Publication.		
Web Resources	1. https://www.programiz.com/python-programming 2. https://www.guru99.com/python-tutorials.html 3. https://www.w3schools.com/python/python_intro.asp 4. https://www.geeksforgeeks.org/python-programming-language/ 5. https://en.wikipedia.org/wiki/Python_(programming_language)		
Pedagogy			
Lecture by Chalk and talk, Power point presentation, E-content, Simulation, Group discussion, Assignment, quiz and Seminar.			
<table border="1" style="width: 100%; border-collapse: collapse;"> <tr> <td style="width: 60%; text-align: center; vertical-align: top;"> Course Designer Name and Designation : Signature : </td><td style="width: 40%; text-align: center; vertical-align: top;"> Verified by BoS Chairman </td></tr> </table>		Course Designer Name and Designation : Signature :	Verified by BoS Chairman
Course Designer Name and Designation : Signature :	Verified by BoS Chairman		

Mapping with Programme Outcomes:

COs	PSO1	PSO2	PSO3	PSO4	PSO5	PO1	PO2	PO3	PO4	PO5
CO1	H	H	H	H	H	H	H	H	M	H
CO2	H	H	H	H	M	H	H	H	H	M
CO3	H	H	H	H	M	M	H	H	H	M
CO4	H	H	H	H	M	H	H	H	H	M
CO5	H	M	H	H	H	H	M	H	H	H

H-High M-Medium L-Low

Core Practical – I	PYTHONPROGRAMMING LAB		
Course code	Total Hours	Contact Hours	Credits
24UCACP01	4	4	3
	Course Objectives		
	The main objectives of this course are		
1	Be able to design and program Python applications.		
2	Be able to create loops and decision statements in Python.		
3	Be able to work with functions and pass arguments in Python.		
4	Be able to build and package Python modules for reusability.		
5	Be able to read and write files in Python.		
CO numbers	Expected Course Outcome: On the successful completion of the course, student will be able to:		
CO1	Demonstrate the understanding of syntax and semantics of PYTHON language	K1 K2 K3 K2 K4 K5 K6	
CO2	Identify the problem and solve using PYTHON programming techniques.		
CO3	Identify suitable programming constructs for problem solving.		
CO4	Analyze various concepts of PYTHON language to solve the problem in an efficient way.		
CO5	Develop a PYTHON program for a given problem and test for its correctness.		
	K1-Foundational Knowledge, K2-Application, K3-Integration, K4-Human Dimension, K5-Caring, K6-Learning How to Learn		
	Course content		
	1. Program using variables, constants, I/O statements in Python. 2. Program using Operators in Python. 3. Program using Conditional Statements. 4. Program using Loops. 5. Program using Jump Statements. 6. Program using Functions. 7. Program using Recursion. 8. Program using Arrays. 9. Program using Strings. 10. Program using Modules. 11. Program using Lists.		

	12. Program using Tuples. 13. Program using Dictionaries. 14. Program for File Handling.
Pedagogy	
Lecture by Chalk and talk, Power point presentation, Projector and Assignment	
Course Designer Name and Designation : Signature :	Verified by BoS Chairman

Mapping with Programme Outcomes:

Cos	PSO1	PSO2	PSO 3	PSO 4	PSO 5	PO1	PO2	PO3	PO4	PO5
CO 1	H	H	H	H	H	H	H	H	H	H
CO 2	H	H	L	H	M	H	H	L	H	M
CO 3	H	H	H	H	M	H	H	H	H	M
CO 4	H	H	H	H	M	H	H	H	H	M
CO 5	H	M	H	H	H	H	M	H	H	H

H-High M-Medium L-Low

SEMESTER - II

SEMESTER - II			
Core - II	OBJECT ORIENTED PROGRAMMING USING C++		
25UCAC02	Total Hours	Contact Hours	Credits
	5	5	5
Course Objectives			
The main objectives of this course are			
1	Describe the procedural and object oriented paradigm with concepts of streams, classes, functions, data and objects		
2	Understand dynamic memory management techniques using pointers, constructors, destructors, etc		
3	Describe the concept of function overloading, operator overloading, virtual functions and polymorphism		
4	Classify inheritance with the understanding of early and late binding, usage of exception handling, generic programming		
5	Demonstrate the use of various OOPs concepts with the help of programs		
CO numbers	Expected Course Outcome: On the successful completion of the course, student will be able to: Develop algorithmic solutions to simple computational problems		
CO1	Remember the program structure of C with its syntax and semantics	K1 K2 K3 K2 K4 K5 K6	
CO2	Understand the programming principles in C(datatypes, operator, branching and looping, arrays, functions, structures, pointers and files)		
CO3	Apply the programming principles learnt in real-Time problems		
CO4	Analyze the various methods of solving a problem And choose the best method		
CO5	Code, debug and test the programs with appropriate test cases		
	K1 -Foundational Knowledge, K2 -Application, K3 -Integration, K4 -Human Dimension, K5 -Caring, K6 -Learning How to Learn		
	Course content		
I	Introduction to C++-key concepts of Object-Oriented Programming–Advantages–Object Oriented Languages–I/O in C++-C++Declarations. Control Structures:-Decision Making and Statements: If..else, jump, goto, break, continue, Switch case statements- Loops in C++: for, while, do-functions in C++-inline functions–Function Overloading.		
II	Classes and Objects: Declaring Objects – Defining Member Functions –Static Member variables and functions–array of objects–friend functions–Overloading member functions–Bit fields and classes–Constructor and destructor with static members.		
III	Operator Overloading: Overloading unary, binary operators–Overloading Friend functions–type conversion–Inheritance: Types of Inheritance–Single, Multilevel, Multiple, Hierarchal, Hybrid, Multipath inheritance–Virtual base Classes–Abstract Classes.		
IV	Pointers–Declaration–Pointer to Class, Object–this pointer–Pointers to derived classes and Base classes–Arrays–Characteristics–array of classes–Memory models–new and delete operators–dynamic object –Binding, Polymorphism and Virtual Functions.		
V	Files –File stream classes –file modes–Sequential Read /Writeoperations–		

	Binary and ASCII Files–Random Access Operation–Templates–Exception Handling - String–Declaring and Initializing string objects–String Attributes–Miscellaneous functions. *Case Study: ❖ Project 1: Create Tic Tac Toe game using C++ concepts using AI (Student must submit the report)
Text Books	1. E. Balagurusamy, –Object-Oriented Programming with C++ II, TMH 2013, 7th Edition.
Reference Books	1. Ashok N Kamthane, –Object-Oriented Programming with ANSI and Turbo C++ II, Pearson Education 2003. 2. Maria Litvin & Gray Litvin, –C++ for you II, Vikas publication 2002.
Web Resources	1. https://alison.com/course/introduction-to-c-plus-plus-programming
Pedagogy	
Lecture by Chalk and talk, Power point presentation, E-content, Simulation, Group discussion, Assignment, quiz and Seminar.	
Course Designer Name and Designation : Signature :	Verified by BoS Chairman

MAPPING WITH PROGRAMME OUTCOMES

Cos	PSO1	PSO2	PSO3	PSO4	PSO5	PO1	PO2	PO3	PO4	PO5
CO1	H	M	L	-	-	H	M	L	-	-
CO2	M	M	M	L	-	M	M	M	L	-
CO3	H	L	L	-	L	H	L	L	-	L
CO4	L	M	L	M	M	L	M	L	M	M
CO5	H	M	L	M	H	H	M	L	M	H

H-High; M-Medium; L-Low

Core Practical – II	C++ PROGRAMMING LAB		
25UCACP02	Total Hours	Contact Hours	Credits
	4	4	3
Course Objectives			
The main objectives of this course are			
1	Describe the procedural and object oriented paradigm with concepts of streams, classes, functions, data and objects		
2	Understand dynamic memory management techniques using pointers, constructors, destructors, etc		
3	Describe the concept of function overloading, operator overloading, virtual functions and polymorphism		
4	Classify inheritance with the understanding of early and late binding, usage of exception handling, generic programming		
5	Demonstrate the use of various OOPs concepts with the help of programs		
CO numbers	Expected Course Outcome: On the successful completion of the course, student will be able to: Write, test, and debug simple Python programs. Read and write data from/to files in Python		
CO1	Remember the program structure of C with its syntax and semantics	K2 K3 K2 K4 K5 K6	
CO2	Understand the programming principles in C (data types, operators, branching and looping, arrays, functions, structures, pointers and files)		
CO3	Apply the programming principles learnt in real-time problems		
CO4	Analyze the various methods of solving a problem and choose the best method		
CO5	Code, debug and test the programs with appropriate test cases		
	K1-Foundational Knowledge, K2-Application, K3-Integration, K4-Human Dimension, K5-Caring, K6-Learning How to Learn		
	Course content		
1.	Write a C++ program to demonstrate function overloading, Default Arguments and Inline function.		
2.	Write a C++ program demonstrate Class and Objects		
3.	Write a C++ program to demonstrate the concept of Passing Objects to Functions		
4.	Write a C++program to demonstrate the Friend Functions.		
5.	Write a C++ program to demonstrate the concept of Passing Objects to Functions		
6.	Write a C++ program to demonstrate Constructor and Destructor		
7.	Write a C++ program to demonstrate Unary Operator Overloading		
8.	Write a C++ program to demonstrate Binary Operator Overloading		
9.	Write a C++ program to demonstrate: - Single Inheritance - Multilevel Inheritance		

	• Multiple Inheritance • Hierarchical Inheritance • Hybrid Inheritance		
10.	Write a C++ program to demonstrate Virtual Functions.		
11.	Write a C++ program to manipulate a Text File.		
12.	Write a C++ program to perform Sequential I/O Operations on a file.		
13.	Write a C++ program to find the Biggest Number using Command Line Arguments		
14.	Write a C++ program to demonstrate Class Template		
15.	Write a C++ program to demonstrate Function Template.		
16.	Write a C++ program to demonstrate Exception Handling.		
Text Books	1. E. Balagurusamy, -Object-Oriented Programming with C++ , TMH2013, 7th Edition.		
Reference Books	1. Ashok NKamthane, -Object-Oriented Programming with ANSI and TurboC++ , Pearson Education 2003. 2. Maria Litvin & Gray Litvin, -C++ for you , Vikas publication 2002.		
Web Resources	1. https://alison.com/course/introduction-to-c-plus-plus-programming		
Pedagogy			
Lecture by Chalk and talk, Power Point Presentation, Projector and Assignment			
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MAPPING WITH PROGRAMME OUTCOMES

Cos	PSO1	PSO2	PSO3	PSO4	PSO5	PO1	PO2	PO3	PO4	PO5
CO1	H	H	H	H	L	H	H	H	H	L
CO2	M	H	H	H	L	M	H	H	H	L
CO3	M	H	H	H	L	M	H	H	H	L
CO4	M	H	H	H	L	M	H	H	H	L
CO5	M	H	H	H	L	M	H	H	H	L

H-High; M-Medium; L-Low

Non Major Elective – (NME)

Subjects offered by the Department of Computer Science

The department offer the subjects to the other major subject students in each semester.

SEMESTER-1

NME	DIGITAL ADVERTISING AND STRATEGIES		
Course code	Total Hours	Contact Hours	Credits
25UCANM1	2	2	2
	Course Objectives		
	The main objectives of this course are		
1	The fundamentals of digital advertising and its components (Knowledge).		
2	social media marketing strategies to create targeted ad campaigns (Application).		
3	The effectiveness of search engine marketing and SEO techniques (Analysis).		
4	The performance of digital marketing campaigns using analytics and key performance indicators (Evaluation).		
5	comprehensive digital advertising strategy by integrating various online marketing tools and techniques (Synthesis).		
CO numbers	Expected Course Outcome: On the successful completion of the course, student will be able to:		
CO1	To define digital advertising concepts and identify key strategies in social media, SEM, SEO, email marketing, and content creation.	K1 K2 K3 K2 K4 K5 K6	
CO 2	Design and execute effective digital advertising campaigns using platforms like Google Ads, Meta Ads Manager, and Mailchimp.		
CO 3	analyze performance metrics from digital campaigns, assess SEO rankings, and recommend improvements based on data-driven insights.		
CO 4	digital marketing strategies, using web analytics and A/B testing to optimize campaigns for maximum engagement and return on investment (ROI).		
CO 5	develop a comprehensive, integrated digital advertising strategy, using insights from various digital tools to drive targeted traffic, improve conversions, and achieve business goals.		
	K1 -Foundational Knowledge, K2 -Application, K3 -Integration, K4 -Human Dimension, K5 -Caring, K6 -Learning How to Learn		
	Course content		
I	Digital Advertising • Definition and scope of digital advertising • Differences between traditional and digital advertising • Overview of online consumer behavior • Key components of a digital marketing strategy Practical Content: • Task: Analyze a digital ad campaign (e.g., on YouTube or Google Ads) and create a summary report •		

II	Social Media Advertising • Introduction to social media platforms: Facebook, Instagram, LinkedIn, Twitter • Creating effective ads for each platform • Target audience segmentation and ad personalization • Budgeting and bidding strategies Practical Content: • Task: Design a mock social media ad campaign for a small business using Meta Ads Manager
III	Search Engine Marketing (SEM) and SEO • Introduction to SEM: Google Ads and Bing Ads • Basics of keyword research and PPC advertising • On-page and off-page SEO techniques • SEO tools: Google Search Console, SEMrush, Yoast Practical Content: • Task: Create a basic Google Ads campaign and conduct keyword research using Google Keyword Planner
IV	Email and Content Marketing • Importance of email marketing in customer retention • Crafting engaging email newsletters • Content strategy and storytelling in advertising • Introduction to content marketing tools: Mailchimp, Canva, HubSpot Practical Content: Task: Design an email marketing campaign for a product launch using Mailchimp
V	Analytics and Advertising Strategy • Web analytics: Google Analytics basics • Campaign performance metrics: CTR, ROI, Conversion Rates • A/B testing and data-driven decision making • Creating a complete digital advertising strategy Practical Content: Task: Analyze a sample campaign report and optimize ad strategy using insights
Text Books	"Digital Marketing for Dummies" by Ryan Deiss and Russ Henneberry, 1st Edition, Wiley Publishing (2017)
Reference Books	1. "Digital Marketing Strategy" by Simon Kingsnorth, 2nd Edition, Kogan Page (2019) 2. "SEO 2023: Learn Search Engine Optimization" by Adam Clarke, 7th Edition, Independently Published (2023)

	3."Email Marketing Rules" by Chad S. White, 3rd Edition, CreateSpace Independent Publishing (2017)
Pedagogy	
Lecture by Chalk and talk, Power point presentation, E-content, Simulation, Group discussion, Assignment, quiz and Seminar.	
Course Designer Name and Designation : Signature :	Verified by BoS Chairman

Cos	PSO 1	PSO2	PSO3	PSO4	PSO5	PO 1	PO2	PO3	PO4	PO5
CO1	H	H	H	H	H	H	H	H	M	H
CO2	H	H	H	H	M	H	H	H	H	M
CO3	H	H	H	H	M	M	H	H	H	M
CO4	H	H	H	H	M	H	H	H	H	M
CO5	H	M	H	H	H	H	M	H	H	H

H-High M-Medium L-Low

SEMESTER-2

NME	DIGITAL SKILLS FOR EMPLOYABILITY		
Course code	Total Hours	Contact Hours	Credits
25UCANM2	2	2	2
	Course Objectives		
	The main objectives of this course are		
1	Essential digital skills needed in the modern workplace.		
2	Equip students with knowledge of productivity tools, communication platforms, and online collaboration.		
3	Develop skills in digital literacy, cybersecurity basics, and online professional presence.		
4	Enhance employability through digital project management and remote working tools.		
5	Prepare students for future technological adaptability and lifelong digital learning.		
CO numbers	Expected Course Outcome: On the successful completion of the course, student will be able to:		
CO1	Understand the importance of digital literacy in today’s job market.	K1 K2 K3 K2 K4 K5 K6	
CO 2	Apply productivity and communication tools for effective work.		
CO 3	Create and manage professional digital profiles and portfolios.		
CO 4	Create and manage professional digital profiles and portfolios.		
CO 5	Evaluate the impact of emerging digital trends on employability.		
	K1-Foundational Knowledge, K2-Application, K3-Integration, K4-Human Dimension, K5-Caring, K6-Learning How to Learn		
	Course content		
I	Digital Literacy - What is Digital Literacy? - Importance of digital skills for employment - Digital Etiquette and Netiquette - Overview of digital workplace environments		
II	Productivity and Collaboration Tools - Word Processing, Spreadsheets, and Presentation Tools (MS Office, Google Workspace) - Cloud Storage and Sharing (Google Drive, OneDrive) - Online Collaboration Platforms (Slack, Trello, Microsoft Teams)		
III	Digital Communication and Professional Networking - Email Communication (writing professional emails) - Video Conferencing tools (Zoom, Google Meet, Microsoft Teams) - Building a Professional Profile (LinkedIn essentials) - Personal Branding Online		

IV	Cybersecurity Basics • Importance of Cybersecurity • Safe Browsing Practices • Password Management • Understanding Phishing and Scams • Introduction to Data Privacy (GDPR basics)
V	Emerging Digital Trends and Employability • Remote Work Skills • Introduction to Artificial Intelligence and Automation • Digital Transformation and Future Work Skills • Preparing for Lifelong Digital Learning
Text Books	1. Digital Skills: Unlock Your Potential for the Digital Economy, Dr. Alan Charlesworthm Routledge ,1st Edition
Reference Books	1.The Essential Digital Skills Handbook Anna B. Harrison Kogan Page Latest Edition 2.Introduction to Information Technology Efraim Turban, R. Kelly Rainer, Richard E. Potter: Wiley3rd Edition 3.Cybersecurity Essentials , Charles J. Brooks, Christopher Grow, Philip Craig, Donald Short , Wiley , 1st Edition
Pedagogy	
Lecture by Chalk and talk, Power point presentation, E-content, Simulation, Group discussion, Assignment, quiz and Seminar.	
Course Designer Name and Designation : Signature :	Verified by BoS Chairman

Cos	PSO 1	PSO2	PSO3	PSO4	PSO5	PO 1	PO2	PO3	PO4	PO5
CO1	H	H	H	H	H	H	H	H	M	H
CO2	H	H	H	H	M	H	H	H	H	M
CO3	H	H	H	H	M	M	H	H	H	M
CO4	H	H	H	H	M	H	H	H	H	M
CO5	H	M	H	H	H	H	M	H	H	H

H-High M-Medium L-Low

SKILL ENHANCEMENT COURSE (SEC)

SEMESTER - I

SEC-I	COMPUTER SCIENCE WITH AI ACCELERATION -FOUNDATIONS		
Course code	Total Hours	Contact Hours	Credits
25UCAS01	4	4	4
Course Objectives			
The main objectives of this course are			
1	To develop foundational knowledge of AI concepts, latest AI tools (ChatGPT, Google Gemini, Claude), and their practical applications in small business contexts.		
2	To apply basic AI tools to solve real-world problems in web development, content creation, and digital marketing for local SMEs.		
3	To integrate AI concepts with traditional business practices and understand their interconnections in the regional business ecosystem.		
4	To recognize the human dimension of AI adoption by exploring how it impacts local businesses, employees, and customers in Erode/Salem/Namakkal regions.		
5	To foster a caring mindset about ethical AI use, data privacy, and responsible technology adoption in small business environments.		
6	To develop self-directed learning strategies that enable continued exploration of emerging AI tools and applications beyond the classroom		
CO numbers	Expected Course Outcome: On the successful completion of the course, student will be able to:		
CO1	Demonstrate foundational knowledge of AI concepts, ChatGPT, Google Gemini, and their practical business applications	K1 K2 K3 K4 K5 K6	
CO2	Apply AI tools to create basic websites, generate content, and solve authentic SME problems		
CO3	Connect AI capabilities with existing business processes to enhance productivity and customer service		
CO4	Examine how AI adoption impacts local business owners, employees, and communities in Tamil Nadu		
CO5	Develop ethical values regarding data privacy, AI transparency, and responsible use in business contexts		
CO6	Create personalized strategies for continuous learning about emerging AI tools and industry applications		
	K1-Foundational Knowledge, K2-Application, K3-Integration, K4-Human Dimension, K5-Caring, K6-Learning How to Learn		
	Course content		

I	AI Fundamentals for Business ● Introduction to AI and Machine Learning – Demystifying concepts for non-technical users ● Latest AI Tools Overview – ChatGPT, Google Gemini, Claude, Perplexity AI ● Practical AI Applications in Tamil Nadu SMEs – Real local case studies ● Setting up and Using AI Tools – Account creation, prompt basics, free vs
	paid versions
II	AI for Content and Communication (15 hours) ● Content Generation with AI – Blog posts, social media content, product descriptions ● Multilingual Content Creation – Tamil and English content strategies ● AI-Powered Email Marketing – Personalization and automation basics ● Customer Communication Enhancement – Chatbots and automated responses
III	AI-Enhanced Web Presence (15 hours) ● AI Tools for Website Creation – Using AI website builders and WordPress integration ● SEO Content Generation – AI-assisted keyword research and content optimization ● Visual Content Creation – AI image generators (DALL-E, Midjourney basics) ● Website Content Management – Using AI to maintain and update websites
IV	AI for Business Operations (15 hours) ● AI in Digital Marketing – Social media scheduling, ad copy generation ● Basic Data Analysis with AI – Understanding business metrics using AI tools
V	Implementation ● AI for Customer Service – Simple chatbot implementation ● Cost-Benefit Analysis – Evaluating AI tool investments for SMEs
Text Books	1. Russell, S. & Norvig, P., Artificial Intelligence: A Modern Approach (4th Edition), Pearson, 2021. 2. Brynjolfsson, E. & McAfee, A., The Second Machine Age, W. W. Norton & Company, 2016. 3. Davenport, T.H., The AI Advantage, MIT Press, 2018. 4. Iansiti, M. & Lakhani, K.R., Competing in the Age of AI, Harvard Business Review Press, 2020. 5. Lee, K.F., AI Superpowers, Houghton Mifflin Harcourt, 2018.

Reference Books	1. Agrawal, A., Gans, J. & Goldfarb, A., Prediction Machines, Harvard Business Review Press, 2018. 2. Daugherty, P.R. & Wilson, H.J., Human + Machine, Harvard Business Review Press, 2018. 3. Burgess, A., The Executive Guide to Artificial Intelligence, Palgrave Macmillan, 2018. 4. Chui, M. & Manyika, J., AI in Business: A McKinsey Global Survey, McKinsey & Company, 2023.
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Web Resources	1. https://chat.openai.com - Official ChatGPT platform for hands-on practice 2. https://gemini.google.com - Google Gemini AI interface for practical exercises 3. https://www.coursera.org/courses?query=ai%20for%20everyone - AI fundamentals courses 4. https://developers.google.com/machine-learning/crash-course - Google's ML basics 5. https://www.myscheme.gov.in/schemes/msme-stes - Tamil Nadu MSME skill resources
Pedagogy	
Hands-on workshops, local SME visits, AI tool demonstrations, collaborative problem-solving, case-based learning with regional businesses, peer teaching sessions, reflective practice journals, and industry mentor interactions.	
Course Designer Name and Designation : Signature :	Verified by BoS Chairman

Mapping with Programme Outcomes:

Cos	PSO 1	PSO2	PSO3	PSO4	PSO5	PO 1	PO2	PO3	PO4	PO5
CO1	H	H	H	H	H	H	H	H	M	H
CO2	H	H	H	H	M	H	H	H	H	M
CO3	H	H	H	H	M	M	H	H	H	M
CO4	H	H	H	H	M	H	H	H	H	M
CO5	H	M	H	H	H	H	M	H	H	H

H-High M-Medium L-Low

SEMESTER – II

SEC-II	COMPUTER SCIENCE WITH AI ACCELERATION - IMPLEMENTATION		
Course code	Total Hours	Contact Hours	Credits
25UCAS02	4	4	4
Course Objectives			
The main objectives of this course are			
1	To develop foundational knowledge of practical AI implementation in web development, mobile apps, and business automation for regional SMEs.		
2	To apply AI-enhanced development tools to create functional solutions addressing real business needs in Erode/Salem/Namakkal markets.		
3	To integrate AI capabilities with existing business systems and understand technical implementation challenges.		
4	To recognize the human dimension of AI implementation by understanding user experience, training needs, and change management.		
5	To foster a caring mindset about sustainable technology adoption and long-term business impact.		
6	To develop self-directed learning strategies for mastering emerging AI development tools and frameworks.		
CO numbers	Expected Course Outcome: On the successful completion of the course, student will be able to:		
CO1	Demonstrate foundational knowledge of AI-powered development tools and implementation frameworks	K1 K2 K3 K4 K5 K6	
CO2	Apply AI tools to develop functional websites, mobile apps, and automation solutions for SMEs		
CO3	Connect AI implementation with business workflows to create integrated solutions		
CO4	Examine how AI implementation affects workplace dynamics and skill requirements in local businesses		
CO5	Develop professional values regarding quality assurance, user privacy, and sustainable tech adoption		
CO6	Create personalized learning paths for advanced AI development skills and industry certifications		
	K1-Foundational Knowledge, K2-Application, K3-Integration, K4-Human Dimension, K5-Caring, K6-Learning How to Learn		
	Course content		

I	AI-Powered Web Development (15 hours) - AI Code Assistants – GitHub Copilot, Cursor AI, Codeium for beginners - Building Responsive Websites – AI-assisted HTML, CSS, JavaScript development - WordPress Enhancement with AI – Custom themes, plugin integration - E-commerce Solutions – WooCommerce setup with AI optimization
II	Mobile App Development with AI (15 hours) - No-Code App Builders – Bubble, FlutterFlow with AI features - Progressive Web Apps – Converting websites to mobile experiences - AI in App Design – User interface generation and optimization - Local Business App Solutions – Inventory, booking, delivery apps
III	Business Automation Implementation (15 hours) - Workflow Automation – Zapier, Make.com for SME processes - AI-Enhanced CRM Systems – HubSpot, Zoho integration basics - Document Processing – AI for invoicing, inventory management - WhatsApp Business API – Automated customer communication
IV	Data Analytics and Insights (15 hours) - Google Analytics with AI Insights – Understanding customer behavior - Sales Data Analysis – Using AI for forecasting and trends
V	AI Insights - Social Media Analytics – AI tools for engagement optimization - ROI Measurement – Tracking AI implementation success
Text Books	- Burkov, A., The Hundred-Page Machine Learning Book, Self-published, 2019. - VanderPlas, J., Python Data Science Handbook, O'Reilly Media, 2023. - McKinney, W., Python for Data Analysis, O'Reilly Media, 2022. - Tamil Nadu IT Exports Report, NASSCOM & State Government, 2024. - SME Digital Transformation Playbook, Ministry of MSME, Government of India, 2024.
Reference Books	- Burkov, A., The Hundred-Page Machine Learning Book, Self-published, 2019. - VanderPlas, J., Python Data Science Handbook, O'Reilly Media, 2023. - McKinney, W., Python for Data Analysis, O'Reilly Media, 2022. - Tamil Nadu IT Exports Report, NASSCOM & State Government, 2024. - SME Digital Transformation Playbook, Ministry of MSME, Government of India, 2024.

	3. Kumar, A., "Digital Transformation for MSMEs in India", Pearson India, Delhi, 2024. 4. Murthy, S., "Data Engineering for AI Applications", Oxford University Press, New Delhi, 2023. 5. Patel, H., "Change Management for Digital Transformation", Orient BlackSwan, Hyderabad, 2022.
Web Resources	1. https://github.com/features/copilot - GitHub Copilot for code assistance 2. https://www.tensorflow.org/js - TensorFlow.js for web-based AI 3. https://teachablemachine.withgoogle.com - Google's Teachable Machine 4. https://www.kaggle.com/learn - Practical AI/ML tutorials 5. https://tnsdc.skillsnetwork.site - Tamil Nadu Skill Development courses
Pedagogy	
Industry-connected project-based learning, implementation case studies from regional businesses, simulations of real-world AI deployments, collaborative problem-solving workshops, experiential learning through prototyping, reflective practice focusing on implementation challenges, and digital portfolios showcasing industry-relevant solutions.	
Course Designer Name and Designation : Signature : Verified by BoS Chairman	

Mapping with Programme Outcomes:

Cos	PSO 1	PSO2	PSO3	PSO4	PSO5	PO 1	PO2	PO3	PO4	PO5
CO1	H	H	H	H	H	H	H	H	M	H
CO2	H	H	H	H	M	H	H	H	H	M
CO3	H	H	H	H	M	M	H	H	H	M
CO4	H	H	H	H	M	H	H	H	H	M
CO5	H	M	H	H	H	H	M	H	H	H

H-High M-Medium L-Low